Continuity equation:

$$\sum F = \dot{m}(v_f - v_i) \qquad \sum T = \dot{m}r(V_f - V_i - r\omega)$$

$$v = r\omega$$

$$\dot{m} = \rho Q = \rho V A$$

Bernoulli's equation (at a point):

$$\frac{p_1}{\rho g} + \frac{V_1^2}{2g} + z_1 = \frac{p_2}{\rho g} + \frac{V_2^2}{2g} + z_2$$

Power to head (height):

$$\dot{W} = \dot{m}gh = \rho g Q h \qquad h = \frac{\dot{W}}{\rho g Q} = \frac{\dot{W}}{\dot{m}g}$$

Energy equation (average of particles):

$$\dot{m}_i\left(\frac{p_i}{\rho} + \frac{V_i}{2} + gz_i\right) = \dot{m}_o\left(\frac{p_o}{\rho} + \frac{V_o}{2} + gz_o\right)$$